

Name _____

Section _____

MTH:2310 DISCRETE MATH

QUIZ 8

DUE: FRIDAY NOV. 14

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

Problem 1. (10 points) *Primes and Greatest Common Divisors*

- (a) (6 points) Find the greatest common divisor of 65 and 231 using the Euclidean algorithm. Show every step.

- (b) (4 points) Find the greatest common divisor and least common multiple of 65 and 231 using their prime factorizations.

Problem 2. (10 points) *Solving Linear Congruences*

Solve the following system of linear congruences using the construction method. Show every step.

$$\begin{cases} x \equiv 2 \pmod{3} \\ x \equiv 1 \pmod{4} \\ x \equiv 2 \pmod{7} \end{cases}$$

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Problem 3. (10 points) *Solving Linear Congruences*

Solve the following system of linear congruences using the substitution method. Show every step.

$$\begin{cases} x \equiv 4 \pmod{5} \\ x \equiv 2 \pmod{7} \\ x \equiv 1 \pmod{9} \end{cases}$$

Problem 4. (6 points) *Basics of Counting*

How many bit strings are there of length at most 5 that end with a 1 (not including the empty string) ? Clearly show enough work to justify your answer.

Problem 5. (6 points) *Basics of Counting*

How many natural numbers are less than or equal to 360 but not a multiple of 9? Clearly show enough work to justify your answer.

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Problem 6. (6 points) *Pigeonhole Principle*

A roller coaster train can seat a total of 36 riders. The roller coaster was tracked for a span of two hours. In that time, the ride was operated 17 times and had 597 riders. Explain why the roller coaster must have been operated at least once at full capacity.